

C-DOT Unveils 14 Indigenous Quantum Products for Secure Communication Networks

Source: TH

Why in News?

The **Centre for Development of Telematics (C-DOT)**, under the **Department of Telecommunications (DoT), Ministry of Communications**, unveiled **14 indigenous quantum products** to strengthen the security of India's communication networks, focusing on **Quantum Key Distribution (QKD)** and **Post-Quantum Cryptography (PQC)** during its **43 rd Foundation Day celebrations**.

What are the Key Quantum Technologies Developed by C-DOT?

- **Quantum Key Distribution (QKD) Systems:**
 - **Q-AKSHAY CD:** A compact fibre-based **Quantum Key Distribution (QKD)** system using **Coherent One-Way (COW)** and **Differential Phase Shift (DPS)** protocols, **used for secure quantum key generation and distribution over fibre networks**.
 - **Q-AKSHAY MD:** A fibre-based **QKD** system using the **Measurement Device Independent (MDI)** protocol, **used for secure key distribution with enhanced protection against measurement-device vulnerabilities**.
- **Quantum-Safe Encryptors:**
 - **Q-VIKRAM & Q-PARAKRAM:** Defence-grade quantum-safe encryptors using **Post-Quantum Cryptography (PQC)** algorithms, operating at **Layer 2/3 (Data Link and Network Layers) of the OSI (Open Systems Interconnection) model**, used to secure strategic and defence communications.
 - **Q-AMOGH:** A high-capacity optical encryptor providing up to **200 Gbps** quantum-safe encryption at Layer 1, **used to secure high-speed optical communication networks**.
 - **Q-MAHASETU:** A commercial-grade quantum-safe encryptor supporting up to **40 Gbps** at Layer 2/3, **used to protect enterprise and critical communication networks**.
 - **Q-SETU:** A quantum-safe encryptor providing up to **80 Mbps** at Layer 3, **used to secure commercial and enterprise data communications**.
 - It incorporates **National Institute of Standards and Technology (NIST) PQC algorithms** to protect data against emerging quantum-era threats.
- **Critical Quantum Hardware Components:**

- **C-SPD:** Single-Photon Detector, used to detect single-photon signals in quantum communication systems.
- **C-RD:** Wideband Radio Frequency (RF) Driver, used to drive intensity and phase modulators in quantum communication systems.
- **End-Point and Access Network Solutions:**

QUANTUM-SAFE COMMUNICATION SOLUTIONS

END-POINT AND ACCESS NETWORK SOLUTIONS

Q-DARSHAN

A quantum-safe Video Internet Protocol (IP) Phone, used to secure voice and video communications.

Q-VACHAN

A quantum-safe in-line node, used to upgrade existing IP phones with quantum-safe security.

Q-VAJRA1000

A quantum-safe access node, used to upgrade existing networks such as Gigabit Passive Optical Network (GPON) and wireless radio systems.

Q-VAAYU

A quantum-safe point-to-point wireless network solution, used to secure wireless communication links.

Q-RAQSHAK

A Quantum-Safe Enterprise Network solution, used to protect enterprise communications and IT infrastructure from quantum-era cyber threats.

OSI (Open Systems Interconnection) Model

- It is a **7-layer framework** that explains how data moves between devices over a network:
 - **Layer 1 - Physical:** Transmits raw data as physical signals.
 - **Layer 2 - Data Link:** Handles communication between directly connected devices.
 - **Layer 3 - Network:** Handles **routing and addressing** of data between networks.
 - **Layer 4 - Transport:** Ensures reliable **end-to-end data delivery** .
 - **Layer 5 - Session:** Establishes and manages communication sessions.
 - **Layer 6 - Presentation:** Handles **data formatting, encryption and compression** .
 - **Layer 7 - Application:** Provides network services directly to applications.

What is Quantum Key Distribution (QKD) and Post-Quantum Cryptography (PQC)?

- **Quantum Computing:** It is an emerging field that uses the **principles of quantum mechanics** to solve certain complex problems beyond the capabilities of classical computers.
 - By harnessing quantum phenomena, quantum computers could perform specific calculations in minutes or hours that would take conventional supercomputers thousands of years.
- **Quantum Key Distribution (QKD) Network:** QKD is a highly secure way to share secret encryption keys between two users using quantum particles (photons). It ensures that communication remains safe from hacking.
 - If anyone tries to intercept the key, the quantum particles get disturbed (due to the observer effect and no-cloning rule), which immediately alerts the users about the intrusion.
- **Post-quantum cryptography (PQC):** It is a type of encryption designed to stay secure against attacks

from powerful quantum computers.

QKD	PQC
Based on quantum mechanics	Based on classical mathematics/cryptography
Securely generates and distributes encryption keys	Provides quantum-resistant encryption algorithms
Generally requires specialised quantum communication infrastructure	Can be implemented on existing communication networks
Example: Q-AKSHAY CD, Q-AKSHAY MD	Example: Q-SETU, Q-MAHASETU, Q-VIKRAM

Centre for Development of Telematics (C-DOT)

- **About:** C-DOT established on **24 th August 1984** by the Government of India as a telecom technology development centre under the **Department of Telecommunications (DoT)** .
 - C-DOT is a premier **indigenous telecom R&D institution** , instrumental in India’s telecom revolution, particularly in **rural telephony and network digitisation** .
- **Major Technologies:** Developed solutions in **Next Generation Networks (NGN), Terabit Routers, XGS-PON, DWDM, Wi-Fi, Network Management Systems (NMS), cybersecurity and telecom software** .
- **IoT/M2M:** Developed the **oneM2M standards-based Common Services Platform (CCSP)** to support **Internet of Things (IoT), Machine-to-Machine (M2M), Smart Cities and Industrial IoT** .
- **Emerging Technologies:** C-DOT is also working on **4G/5G, Artificial Intelligence (AI), quantum communication and 6G** technologies.
- **Technology Transfer:** Its **Technology Transfer model** has helped create a domestic ecosystem of **telecom equipment manufacturers and component suppliers** .
- **Significance:** C-DOT advances **Atmanirbhar Bharat, Digital India and indigenous telecom manufacturing** by developing technologies suited to India’s needs.
 - Its technologies have contributed to **BharatNet** , the national optical fibre network aimed at expanding broadband connectivity to rural areas.

Frequently Asked Questions (FAQs)

1. **What is Quantum Key Distribution (QKD)?**

QKD uses **quantum-mechanical principles** to securely generate and distribute encryption keys, with interception detectable through changes in quantum states.

2. **What is Post-Quantum Cryptography (PQC)?**

PQC uses **classical cryptographic algorithms** designed to remain secure against attacks by future quantum computers.

3. **What is the difference between QKD and PQC?**

QKD uses quantum communication infrastructure for secure key distribution , while **PQC uses quantum-resistant mathematical algorithms** that can generally work over existing networks.

4. **What are Q-AKSHAY CD and Q-AKSHAY MD?**

Both are **fibre-based QKD systems** developed by C-DOT. Q-AKSHAY CD uses **COW and DPS protocols** , while Q-AKSHAY MD uses the **Measurement Device Independent (MDI) protocol** .

5. **Why are C-DOT's indigenous quantum products significant for India?**

They strengthen **Atmanirbhar Bharat, cybersecurity and strategic autonomy** by providing indigenous quantum-safe solutions for **defence, telecom, enterprise and critical communication networks** .

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/WOxz5mSHOKk>

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UPSC Civil Services Examination Previous Year Question (PYQ)

Prelims

Q. 'X', born in the UK, was conferred the Nobel Prize in 2025. He was a professor in an American university when this prize was announced. Identify 'X':(2026)

- (a) Michel H. Devoret
- (b) Richard Robson
- (c) John Clarke
- (d) Joel Mokyr

Ans: (C)

Q. Which one of the following is the context in which the term "qubit" is mentioned? (2022)

- (a)** Cloud Services
- (b)** Quantum Computing
- (c)** Visible Light Communication Technologies
- (d)** Wireless Communication Technologies

Ans: (b)

Mains

Q. "The emergence of the Fourth Industrial Revolution (Digital Revolution) has initiated e-Governance as an integral part of government". Discuss. (2020)

India's 1st "Green Forge" Complex Inaugurated

Source: PIB

India's first-of-its-kind **Green Forge Complex** was inaugurated at **National Agri-food & Biomanufacturing Institute (BRIC-NABI)** (an Autonomous Institute of the Department of Biotechnology) in **Mohali**, marking a major step in **agricultural biotechnology** and **genetically modified (GM) crop research**.

- **Context of the Launch:** The facility was inaugurated during the **National Conclave on BioE3 (Biotechnology for Economy, Environment and Employment), 2026** reflecting India's shift toward high-performance biomanufacturing.
- **'Green Forge' Facility:** It provides **advanced simulation chambers** that can precisely regulate light, humidity, and temperature.
 - This allows researchers to cultivate and experiment with **GM crops** to assess their suitability across varied and controlled environmental conditions before field deployment.
- **Technological Integration:** The facility highlights the convergence of biotechnology with **advanced simulation and Artificial Intelligence (AI)**, accelerating complex research in both agricultural and medical sciences.
- **Surge in India's Bioeconomy:** India's **bioeconomy** has expanded from USD 10 billion in 2014 to approximately USD 200 billion, driven by private sector engagement and an enabling policy environment.
- **Infrastructure Push:** Under **BioE3 (Biotechnology for Economy, Environment and Employment) 2024 policy**, the focus is shifting towards **industry-ready manufacturing** and translating research into products and technologies.
 - The government is developing **22 biomanufacturing hubs** and a **National Bio-Foundry Network comprising eight specialized facilities** to support start-ups, SMEs, and the translation of research into commercial products.
 - The **Department of Biotechnology (DBT)** is actively supporting around **38 Carbon Capture and Utilisation (CCU) projects** to capture emissions from hard-to-abate sectors like cement and steel through integrated biomanufacturing.
- **Emerging Biotech Frontiers:** The initiative is expanding into **sustainable aviation fuels (SAF)**, the I3 PhD programme, and **Space Biotechnology** —including collaborations with **ISRO on myogenesis and cyanobacteria** growth experiments in space.
- **Waste-to-Wealth:** BRIC-NABI is heavily integrating agri-biotechnology with bioprocess engineering to convert **agricultural waste into high-value nutraceuticals** (products derived from food sources that provide extra health and medical benefits besides basic nutrition) and **biomaterials**.
- **Strategic Alignment:** The overall transformation toward a low-carbon, circular bioeconomy is directly aligned with India's long-term climate and economic goals of achieving **Net Zero by 2070** and **Atmanirbhar Bharat**.

Read more: Unlocking India's Bioeconomy Potential

